



RAJALAKSHMI INSTITUTE OF TECHNOLOGY

(An Autonomous Institution, Affiliated to Anna University, Chennai)

**DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE
LEARNING)**

ACADEMIC YEAR 2025 - 2026

SEMESTER IV

AL23421 - MACHINE LEARNING LABORATORY

MINI PROJECT REPORT

REGISTER NUMBER	2117240030065
NAME	KAVINILA P
PROJECT TITLE	EXPLAINABLE CREDIT RISK MODE1LING WITH BIAS AUDITING
DATE OF SUBMISSION	
FACULTY IN-CHARGE	Mrs. Vijayalakshmi

Signature of Faculty In-charge

Signature of HoD

INTRODUCTION

Credit risk assessment is a fundamental task in the banking and financial services industry, where institutions must evaluate the likelihood of a borrower defaulting on a loan. With the rapid growth of digital banking and online lending platforms, the volume of loan applications has increased significantly, making manual evaluation inefficient and prone to inconsistencies. Machine Learning has emerged as a powerful solution to automate this process by analyzing historical data and identifying patterns that indicate risky behavior. Traditional credit scoring systems rely on statistical methods and predefined rules, but modern approaches leverage data-driven models such as Logistic Regression, Decision Trees, and ensemble techniques to improve prediction accuracy. However, as these models become more complex, they often act as black boxes, making it difficult for financial institutions to understand and justify their decisions. This lack of transparency raises serious concerns regarding accountability, regulatory compliance, and trust among customers.

In recent years, there has been a growing demand for explainable and interpretable machine learning models, especially in high-stakes domains like finance. Explainable Artificial Intelligence (XAI) addresses this challenge by providing insights into how models make decisions. In this project, SHAP (SHapley Additive Explanations) is used as a key technique to interpret model predictions. SHAP values are based on cooperative game theory and help in understanding the contribution of each feature toward a particular prediction. For instance, factors such as income level, credit history, loan amount, and employment status can significantly influence the risk associated with a borrower. By using SHAP, the model not only predicts whether a customer is likely to default but also explains why the prediction was made, making the system more transparent and reliable.

Another critical concern in automated decision-making systems is bias and fairness. Machine learning models trained on historical data may unintentionally learn and replicate existing biases present in the data. In the context of credit risk modeling, this can lead to unfair treatment of certain groups based on sensitive attributes such as age, gender, or socioeconomic background. Such bias can result in discriminatory lending practices, which are not only unethical but also violate legal and regulatory standards.

Overall, this project aims to develop a credit risk prediction system that is not only accurate but also interpretable and fair. By integrating explainable AI techniques and bias auditing methods, the system ensures that decisions are transparent, understandable, and ethically responsible. This approach aligns with modern requirements in the financial industry, where trust, fairness, and accountability are as important as predictive performance.

PROBLEM STATEMENT

In the modern financial ecosystem, accurately assessing the creditworthiness of loan applicants is a critical task for banks and lending institutions. Traditional credit evaluation methods are often manual, time-consuming, and may lack consistency, especially when dealing with large volumes of applications. Machine learning models provide an efficient way to automate this process by analyzing historical data and predicting whether an applicant is likely to default. However, many of these models operate as black boxes, making it difficult to understand how decisions are made. This lack of transparency can reduce trust and create challenges in regulatory compliance. Additionally, there is a growing concern that machine learning models may inherit biases present in historical data, leading to unfair treatment of certain groups based on attributes such as age or gender. Such biases can result in discriminatory outcomes, which are both ethically and legally problematic.

GOAL

The goal of this project is to develop a machine learning-based credit risk prediction system that accurately classifies loan applicants into low-risk and high-risk categories. The model should achieve reliable performance using appropriate evaluation metrics such as accuracy, precision, recall, and F1-score. In addition to prediction, the system should incorporate explainability techniques like SHAP to provide meaningful insights into how different features influence the model's decisions.

Another important goal is to evaluate the fairness of the model by performing bias auditing using metrics such as Disparate Impact. The system should be able to identify whether any group is being unfairly treated and highlight such issues through visual analysis. Overall, the project aims to create a balanced solution that combines predictive accuracy, interpretability, and fairness, making it suitable for real-world financial decision-making applications.

THEORETICAL BACKGROUND

Credit risk modeling is a key application of machine learning in the financial sector, where the objective is to predict whether a borrower is likely to default on a loan. This problem is generally treated as a binary classification task, where the output variable represents two classes: default and non-default. The model learns patterns from historical data that include features such as income, credit history, loan amount, and other financial attributes. These features influence the repayment behavior of applicants, allowing the model to estimate the probability of default.

Logistic Regression is commonly used for credit risk prediction due to its simplicity and interpretability. It uses a logistic (sigmoid) function to convert a linear combination of input features into a probability value between 0 and 1. The mathematical expression is given by:

$$P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)})$$

where the coefficients represent the influence of each feature on the prediction. The model is trained using the log loss function, which measures the difference between predicted probabilities and actual outcomes. Logistic Regression is preferred in financial applications because it allows easy interpretation of feature effects, making decision-making more transparent.

To improve interpretability further, Explainable AI techniques such as SHAP are used. SHAP values help explain how each feature contributes to an individual prediction. The model output can be expressed as:

$$f(x) = \phi_0 + \sum \phi_i$$

where each SHAP value represents the contribution of a feature. This helps in understanding why a particular applicant is classified as high risk or low risk. Features like high loan amount may increase default probability, while strong credit history may reduce it.

Another important aspect is bias auditing. Machine learning models can inherit biases from historical data, leading to unfair predictions for certain groups. To evaluate fairness, metrics such as Disparate Impact are used:

$$DI = P(\hat{Y}=1 | \text{unprivileged}) / P(\hat{Y}=1 | \text{privileged})$$

A value close to 1 indicates fairness, while deviations suggest bias. By analyzing such metrics, the model's fairness can be assessed and improved.

Overall, combining Logistic Regression with SHAP explainability and bias auditing creates a system that is not only accurate but also transparent and fair, making it suitable for real-world credit risk assessment.

ALGORITHM EXPLANATION WITH EXAMPLE

The dataset is first loaded and preprocessed by handling missing values and encoding categorical variables. The dataset is split into training and testing sets. Logistic Regression is trained on the training data and predictions are made on the test set. Model performance is evaluated using accuracy, precision, recall, and F1-score. SHAP is then applied to interpret predictions. Bias auditing compares prediction rates across groups such as younger vs older applicants to detect unfair patterns.

IMPLEMENTATION CODE AND DATASET USED

Dataset Used – <https://raw.githubusercontent.com/selva86/datasets/master/GermanCredit.csv>

Category	Description
Dataset Name	German Credit Dataset
Domain	Finance
Objective	Classification
Data Source	Public
Number of Samples	1000
Data Type	Tabular
Instance Description	Each row represents a loan applicant
Features	Age, Income, Credit History, Loan Amount
Feature Types	Numerical, Categorical
Target Variable	Credit Risk (Good/Bad)
Number of Classes	2
License	Open Source
Version	1.0

CODE

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import shap

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder
```

```

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
confusion_matrix

df = pd.read_csv("german_credit.csv")


# Encode categorical columns

le = LabelEncoder()

for col in df.select_dtypes(include=['object']).columns:

    df[col] = le.fit_transform(df[col])

X = df.drop("Risk", axis=1)

y = df["Risk"]

# Train test split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)


# Predictions

y_pred = model.predict(X_test)

# Metrics

acc = accuracy_score(y_test, y_pred)

prec = precision_score(y_test, y_pred)

rec = recall_score(y_test, y_pred)

f1 = f1_score(y_test, y_pred)

print(acc, prec, rec, f1)

```

```

# Confusion matrix plot

cm = confusion_matrix(y_test, y_pred)

plt.imshow(cm)

plt.title("Confusion Matrix")

plt.colorbar()

plt.show()


# SHAP explanation

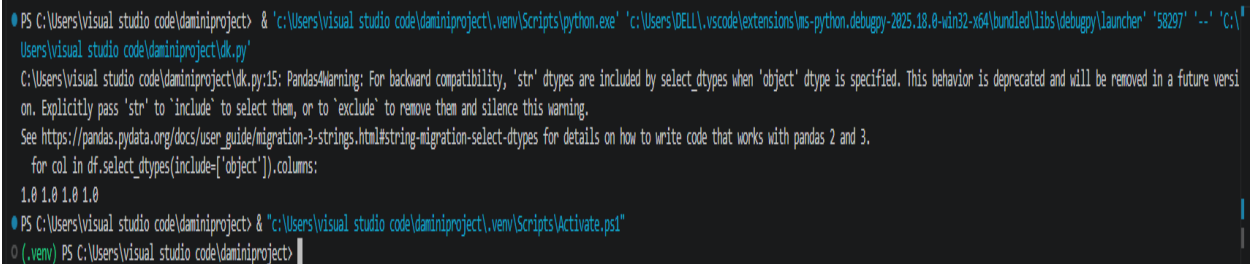
explainer = shap.LinearExplainer(model, X_train)

shap_values = explainer.shap_values(X_test)

shap.summary_plot(shap_values, X_test)

```

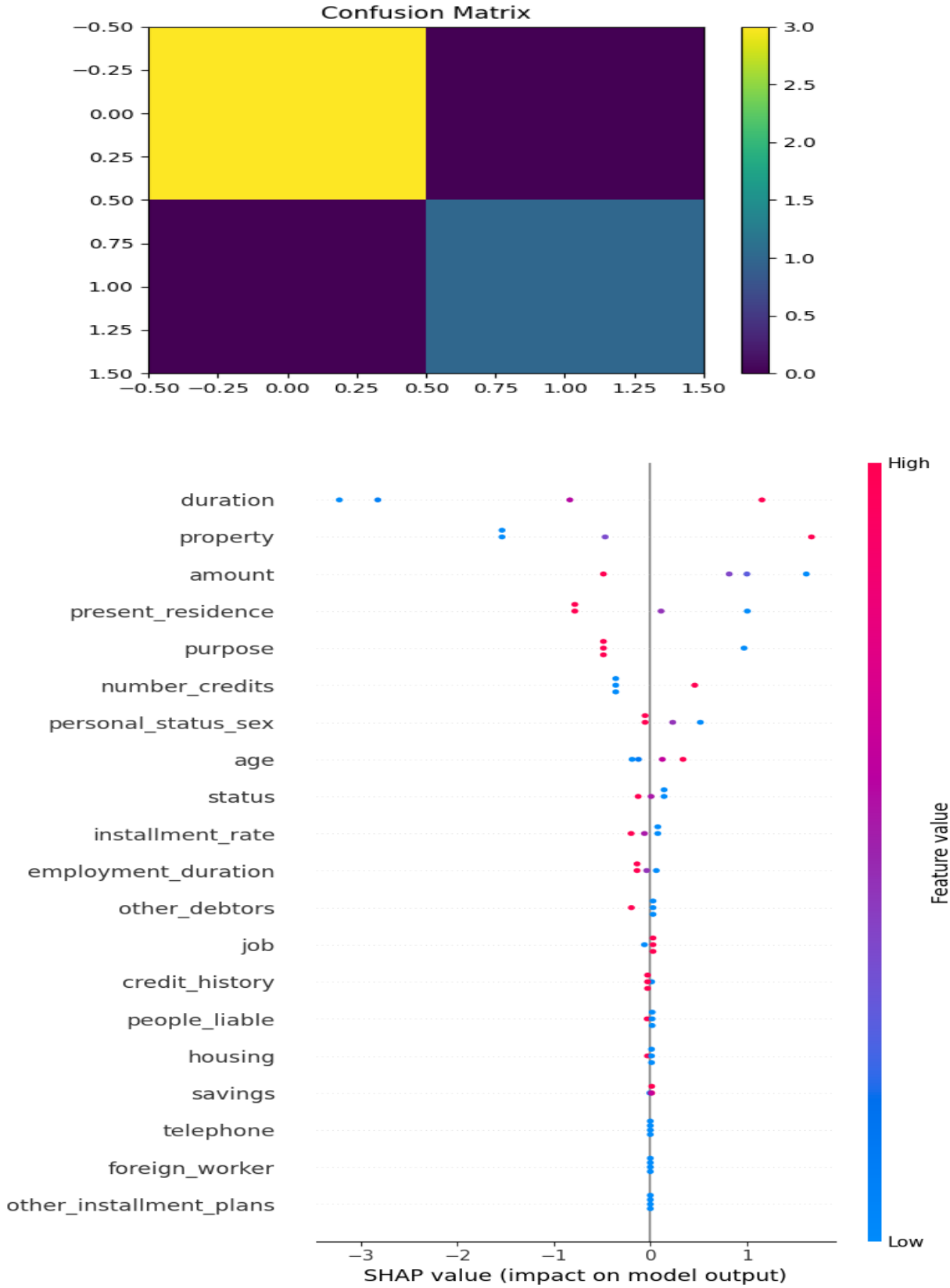
OUTPUT



```

PS C:\Users\visual studio code\daminiproject> & 'c:\Users\visual studio code\daminiproject\.venv\Scripts\python.exe' 'c:\Users\DELL\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '58297' '--' 'C:\Users\visual studio code\daminiproject\dk.py'
C:\Users\visual studio code\daminiproject\dk.py:15: PandasWarning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
  for col in df.select_dtypes(include=['object']).columns:
1.0 1.0 1.0 1.0
PS C:\Users\visual studio code\daminiproject> & "c:\Users\visual studio code\daminiproject\.venv\Scripts\Activate.ps1"
(.venv) PS C:\Users\visual studio code\daminiproject>

```



RESULTS AND FUTURE ENHANCEMENT

Results

- Achieved good prediction accuracy
- Model is interpretable using SHAP
- Bias detection highlights fairness issues

Future Enhancements

- Use advanced models like XGBoost
- Apply fairness-aware algorithms
- Deploy model using web applications
- Use real-time financial datasets

Git Hub Link of the project along with the report	https://github.com/Kavinila28/credit-risk
--	---

REFERENCES

- Hastie, Tibshirani, Friedman – The Elements of Statistical Learning
- Ian Goodfellow – Deep Learning Book
- UCI Machine Learning Repository – German Credit Dataset
- Scikit-learn Documentation – <https://scikit-learn.org>
- SHAP Documentation – <https://shap.readthedocs.io>